

ClassSaga - A Gamified Classroom Incentive System

COMP 596
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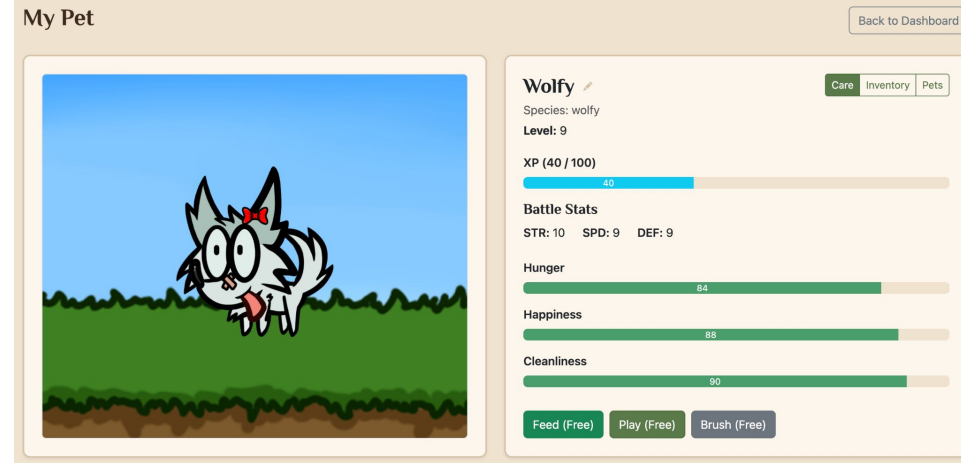
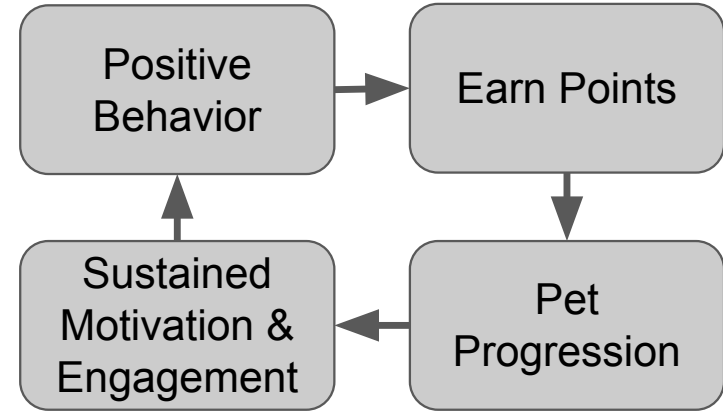
Problem to Solve

- Students struggle with motivation, engagement, and consistent academic participation
- Existing systems:
 - behavior tracking or simple reward economies
 - lack progression and personalization
 - can be difficult to maintain consistently
- **Need for a lightweight, student-centered approach to positive reinforcement**

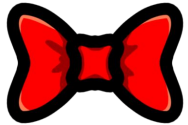


Proposed Solution

- Gamified classroom incentive platform
- Rewards for positive classroom behaviors
- Points for care and customization of pet
- Progression system encourages long-term engagement
- short, intentional sessions



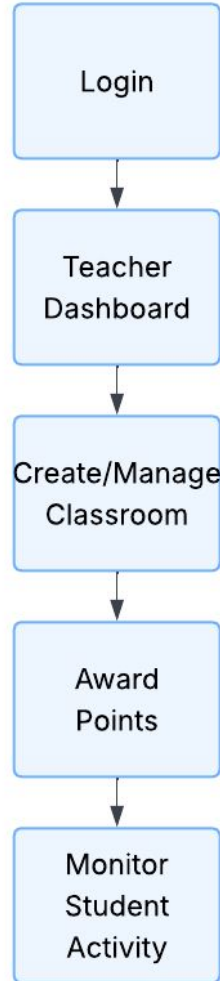
Key Features



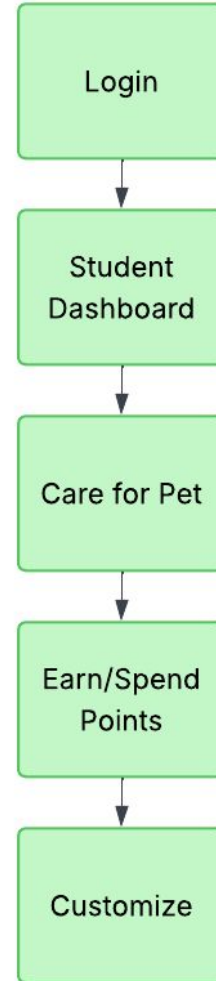
Classroom Management	Virtual Pet System
• Teacher created classrooms • Student enrollment & management • Individual, group, or classwide points	• Started pet selection • Pet care • Level progression and stat growth
Economy & Customization	Activity & Feedback
• Point-based reward economy • Shop with consumables, cosmetics, and pets • Inventory management and equipment	• Real time interaction feedback • Pet animations • Student and classroom activity tracking



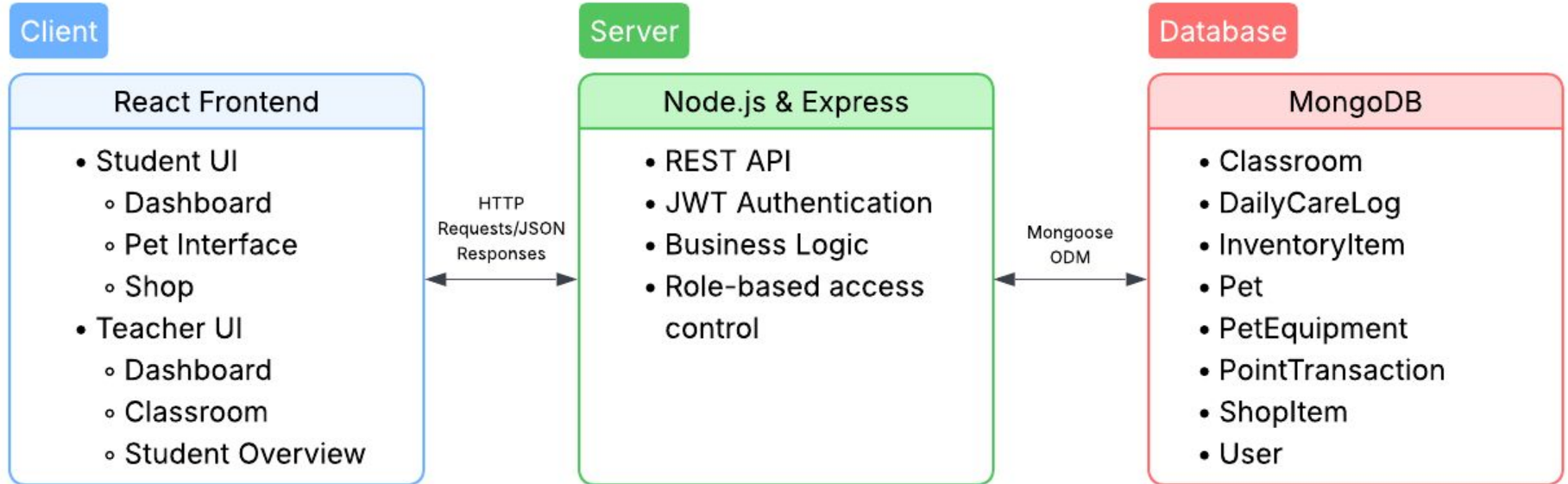
Teacher User Flow



Student User Flow

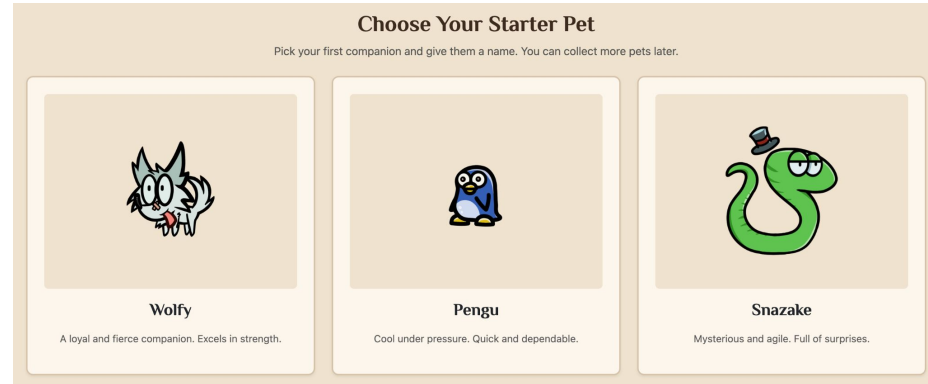


System Overview



Key Design Decisions

- Flexible Data Modeling
 - Separate pets, inventory, shop items, and transactions
- Lightweight Gamification
 - Keep interactions short and low complexity
- Positive Reinforcement & Student Autonomy
 - Rewards engagement rather than penalizing disengagement



Known Limitations

- No LMS Integration
 - Manual student and point management
- Limited Persistent Game State
 - Pet activity log locally stored
 - No active gameplay use for battle stats yet
- No admin interface
- Limited Mobile Optimization

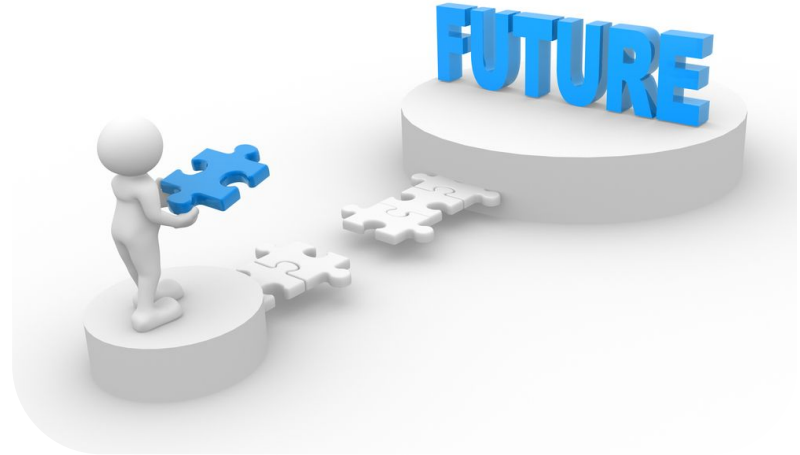


Live Demo



Future Improvements

- NPC battle system using pet battle stats
- Platform integration with Canvas and Google Classroom
- Achievements
- Pet evolution system
- Daily quest/reward
- Multiplayer (whole class) events
- Mobile-optimized layout



Conclusion

- Applies gamification to classroom motivation through a virtual pet reward system
- Combines educational goals with full-stack application design
- Architecture supports future feature expansion and continued development



Questions?

Thank you for your time!